

EDB SUPPORT PACKAGE FOR AI COE · RESEARCH INCENTIVE SCHEME (RIS(C))

EDB Grant Claim Automation

Manpower claims — prepared, verified, audit-ready

ST Engineering · Human Resources

Application No. S26-10249-RIS(C) | Qualifying Period 01 Jan 2026 – 31 Dec 2028 | 17 participating entities

Grant claims are high-stakes and manual

- ▶ **Scale.** 17 participating entities, a three-year qualifying period, and a grant worth up to S\$42 million — manpower only.
- ▶ **Manual & error-prone.** Each claim is hand-built from timesheets, payroll registers and ECMF lists; one wrong rate or a missed exclusion can sink it.
- ▶ **Heavily audited.** An external Public Accountant verifies under SSRS 4400 and samples at least 85% of the claimed value — every figure must trace to a source document.
- ▶ **Hard deadlines.** Per-entity audits from Oct 2026, group submission to EDB by end Nov 2026, final audited claim within 183 days of the period end.

An HR-first portal that does the heavy lifting

- ▶ **Upload, and it takes over.** Drop the HR records; the system reads them, applies EDB's rules, and prepares the claim.
- ▶ **End to end.** Eligibility checks → per-person calculation → the EDB submission pack — in one guided flow.
- ▶ **Private by design.** Runs entirely on your machine; salary data never leaves the building.
- ▶ **Deterministic.** The same inputs always produce identical outputs — a hard audit requirement.

EDB Grant Compliance
ST Engineering - HR - S26-10249-RIS(C)

Claim 01 Jan 2026 - 30 Jun 2026
Support rate confirmed

Simple Advanced

Start a new EDB claim

Add your documents on the right — start with the EDB output template and the trainee list, then the timesheet, ECMF list and payroll. The tracker on the left updates as each is recognised and flags what's still missing.

🛡️ Documents tracked

Upload on the right — recognised files appear here.

REQUIRED — IN ORDER

1

EDB output template

needed

2

Trainee list

needed

3

Team timesheet

needed

4

ECMF researcher list

needed

5

Payroll / payslips

needed

SUPPORTING EVIDENCE

0 added

Any supporting docs you add (leave report, CPF/bank, progress reports...) are recognised and listed here.

📁

Drop your documents here

or click to browse · Excel (.xlsx)

Analyse documents →

Add at least one timesheet workbook to continue.

or try it with sample data (2 companies · 20 staff)

▶ What format does each document need?

🔒 Your files stay on this machine — nothing is uploaded to the internet.

The advantages at a glance

- ▶ **Faster.** Weeks of manual spreadsheet work per claim become minutes — upload to audit-ready pack.
- ▶ **Accurate.** Rules applied the same way every time; reconciles to the cent and removes rate, cap and exclusion mistakes.
- ▶ **Audit-ready.** Every figure is one click from its source cell — a smoother, faster, lower-cost SSRS 4400 audit.
- ▶ **Compliant.** The 60% rate, 9-month cap, S\$42m ceiling and reporting deadlines are enforced, not remembered.
- ▶ **Transparent.** No black box: full workings are shown and nobody is silently dropped — every decision has a reason.
- ▶ **Secure & scalable.** Runs 100% on-premise (salary data never leaves), and applies one consistent method across all 17 entities.

Before vs after

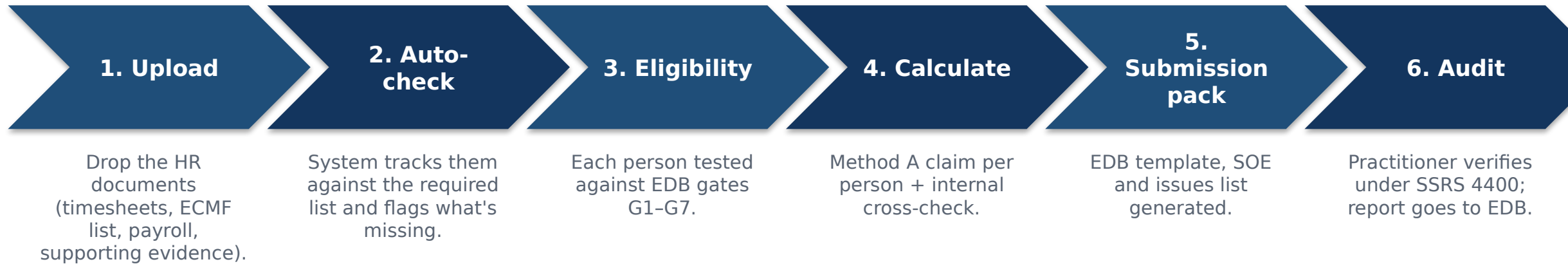
Manual today

- ▶ Weeks of spreadsheet work per claim
- ▶ Easy to mis-apply the rate, cap or an exclusion
- ▶ Audit prep: hunting for source documents by hand
- ▶ Hard to keep 17 entities consistent
- ▶ A rejected claim risks delay or clawback
- ▶ Salary data scattered across files

With this system

- ▶ Minutes: upload → audit-ready submission pack
- ▶ Rules enforced automatically; cent-exact
- ▶ Every figure one click from its source cell
- ▶ One consistent method across all 17 entities
- ▶ Defensible, traceable, deterministic outputs
- ▶ All processing on-premise — nothing leaves

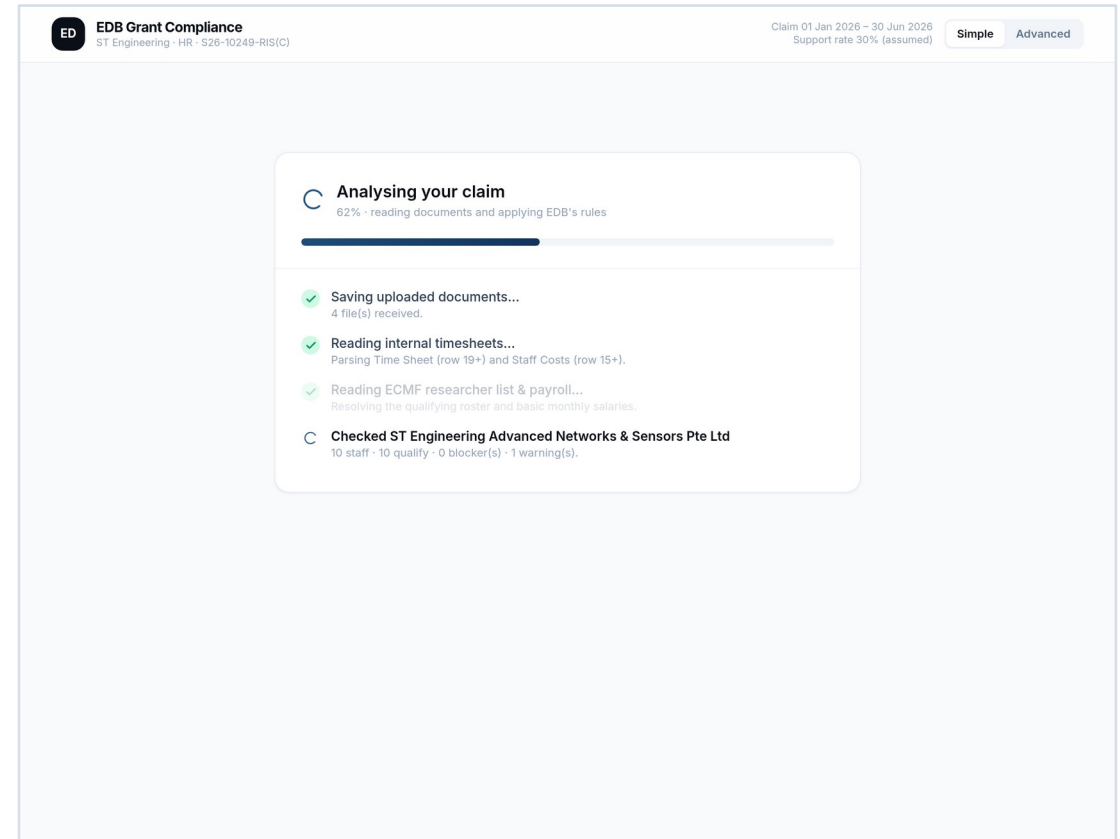
How a claim flows through the system



Upload once → the system checks, calculates and assembles the audit-ready pack. HR stays in control; every figure is traceable and nobody is silently dropped.

Five guided, transparent steps

- ▶ **Documents.** Upload timesheets, the ECMF list and payroll; tick the supporting evidence.
- ▶ **Document check.** Missing-document validation per company and per person.
- ▶ **Eligibility.** Every person checked against EDB's criteria.
- ▶ **Claim amount.** Each claim calculated and shown in full.
- ▶ **Submission pack.** The audit-ready outputs, generated.
- ▶ **Verbose & re-runnable.** You see exactly what it's doing; fix a document and re-run.



The AI assists — the maths is deterministic

- ▶ **The model never computes a claim figure.** All arithmetic is pure, auditable Python. The AI only reads documents and explains.
- ▶ **Full traceability.** Every number links to its exact source cell — payslip, timesheet, ECMF list.
- ▶ **Nobody is silently dropped.** Excluded and blocked people are always listed with the reason.
- ▶ **Proven.** 115 automated tests; reconciles to the cent against EDB's own worked example.

Seven EDB gates, applied to every person

- **The checks.** Citizen/PR · ECMF-validated · no other grant · salary $\geq$ S\$5,000 · eligible R&D role · active in period · payslip present.
- **Clear verdicts.** Qualifies · Not eligible · Needs a document — with the reason behind each.
- **Human in the loop.** Borderline roles are flagged for HR to confirm, not auto-decided.

EDB Grant Compliance
ST Engineering - HR - 925-9249-8832

Claim 01 Jan 2026 - 30 Jun 2026
Support rate confirmed

Simple Advanced **New claim**

Claimable 13 +1 eligible, \$0 (no hours)	Needs a document 1	Not eligible 5	Total claim \$492,934.55
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Document check Eligibility Claim amount Grant & compliance Submission pack

Every person is checked against EDB's criteria. Nobody is silently dropped — those not claimed are shown with the reason.

All	Qualifying	Not claimed	Needs review
Lin Jia Hao ANS-001 - AI Research Engineer	Qualifies		\$34,200.00
Nurul Aisyah Binte Hassan ANS-002 - Machine Learning Engineer	Qualifies		\$17,454.55
Tan Wei Ming ANS-003 - Senior Data Scientist	Qualifies		\$19,000.00
Rajesh Kumar Pillai ANS-004 - Principal AI Scientist	Qualifies		\$72,000.00
Chua Mei Ling ANS-005 - AI Engineer	Eligible - \$0 — review		\$0.00
Goh Boon Keng ANS-006 - Computer Vision Engineer	Qualifies		\$36,000.00
Simon Tay Chin Hock ANS-007 - Engineering Operations Manager	review	Qualifies	\$43,200.00
Lee Hui Shan ANS-008 - AI Research Scientist	Qualifies		\$50,400.00
Mohammed Iqbal Bin Yusoff ANS-009 - Robotics Engineer	Qualifies		\$30,600.00
Cheryl Teo Xin Yi ANS-010 - Deep Learning Engineer	Qualifies		\$41,400.00
Faridah Binte Omar DSO-001 - Junior AI Engineer	Not eligible		
Arjun Mehta DSO-002 - AI Research Engineer	Not eligible		
Wong Kah Wai DSO-003 - Data Engineer	Not eligible		
Priya Nair DSO-004 - NLP Engineer	Not eligible		
Kelvin Ong Wei Sheng DSO-005	Not eligible		
Siti Nurhaliza Rahman DSO-006 - ML/DL Engineer	Needs a document		
Daniel Koh Jun Wei DSO-007 - AI Platform Engineer	Qualifies		\$33,120.00
Anitha Devi Krishnan DSO-008 - Data Scientist	Qualifies		\$38,880.00
Bryan Sim Hong Yi DSO-009 - AI Software Engineer	Qualifies		\$28,080.00
Ng Pei Shan DSO-010 - Senior ML Engineer	Qualifies		\$48,600.00

Ask

Every claim, shown in full

- **EDB monthly pro-ration (Method A).** Capped salary × month involved × time on project, summed × the support rate. This is the figure submitted.
- **Open any person.** See the month-by-month working, each input linked to its source document.
- **Silent cross-check (Method B).** An internal hours-based method runs in the background; differences are flagged, never hidden.

EDB Grant Compliance
ST Engineering HR (201-10249-8552)

Claim 01 Jan 2028 - 30 Jun 2028
Support rate confirmed

Simple Advanced [New claim](#)

Claimable: 13 (+1 eligible, \$0 per hour) Needs a document: 1 Not eligible: 5 Total claim: \$492,934.55

Document check Eligibility Claim amount Grant & compliance Submission pack

Calculated for the 13 qualifying staff using EDB's monthly pro-ration. You don't enter any numbers — open a person to see exactly how their claim is worked out, tracked to each source document.

A EDB method — the figure we submit
For each month: 60% of the capped salary, scaled by how much of the month the person worked and how much of their time was on the project. Add the months up. This is what goes to EDB.

B Internal check — never submitted
Your team's existing hours-based method (project hours ÷ total capacity). We run it quarterly as a second opinion. If A and B differ a lot it usually means a data issue to fix. "Cross-check Q4" — they agree, "Review" — worth a look.

EMPLOYEE	NAME	MONTHLY SALARY	CLAIM AMOUNT	CROSS-CHECK
ANS-001	Lim Jia Hao	\$9,500.00	\$34,200.00	Review
For each month: $\text{capped salary} \times \text{month worked} \times \text{time on project}$. The claim is the sum across months ÷ the 60% support rate. Salary is capped at \$20,000.00/monthly, time on project at 100%. Every figure links to its source document.				
MONTH	CAPPED BASIC SALARY	MONTH WORKED	TIME ON PROJECT	QUALIFYING COST
Jan 2028	\$9,500.00 Q. Payroll	100% Full month	100% Hours ÷ capacity → clamped Q. Time Sheet	\$9,500.00
Feb 2028	\$9,500.00 Q. Payroll	100% Full month	100% Hours ÷ capacity → clamped Q. Time Sheet	\$9,500.00
Mar 2028	\$9,500.00 Q. Payroll	100% Full month	100% Hours ÷ capacity → clamped Q. Time Sheet	\$9,500.00
Apr 2028	\$9,500.00 Q. Payroll	100% Full month	100% Hours ÷ capacity → clamped Q. Time Sheet	\$9,500.00
May 2028	\$9,500.00 Q. Payroll	100% Full month	100% Hours ÷ capacity → clamped Q. Time Sheet	\$9,500.00
Jun 2028	\$9,500.00 Q. Payroll	100% Full month	100% Hours ÷ capacity → clamped Q. Time Sheet	\$9,500.00
Total qualifying cost				\$57,000.00
$\$57,000.00 \times 60\% \text{ support rate} = \$34,200.00$				EDB Method A — the submitted figure
ANS-002	Nurul Aisyah Binte Hassan	\$8,000.00	\$17,454.55	Review
ANS-003	Tan Wei Ming	\$7,000.00	\$19,000.00	Review
ANS-004	Rajesh Kumar Pillai	\$23,000.00	\$72,000.00	Review
ANS-006	Goh Boon Kang	\$10,000.00	\$36,000.00	Review
ANS-007	Simon Tay Chin Hock	\$12,000.00	\$43,200.00	Review
ANS-008	Lee Hui Shan	\$14,000.00	\$50,400.00	Review
ANS-009	Mohammed Iqbal Bin Yusof	\$8,500.00	\$30,600.00	Review
ANS-010	Cheryl Teo Xin Yi	\$11,500.00	\$41,400.00	Review
DSG-007	Daniel Koh Jun Wei	\$9,200.00	\$33,120.00	Review
DSG-008	Anitha Devi Krishnan	\$10,800.00	\$38,880.00	Review
DSG-009	Bryan Sim Hong Yi	\$7,800.00	\$28,080.00	Review
DSG-010	Ng Pei Shan	\$13,500.00	\$48,600.00	Review
Total claim (60% support rate)			\$492,934.55	

Why is there a second "Internal" figure? (cross-check)

The support package, enforced automatically

- ▶ **Support rate 60%** of the qualifying monthly salary (EDB Support Package).
- ▶ **Basic salary only.** No CPF, bonus, AWS, allowances, COLA or airfare.
- ▶ **Floor S\$5,000 · Cap S\$20,000/month** → at most S\$12,000 funded per person per month.
- ▶ **New hire vs upskill.** New hires funded across the qualifying period; upskilling to PL3 funded for up to 9 months.
- ▶ **Ceiling S\$42m, claim $\geq$ 3 months.** Manpower-only grant cap, with a minimum claim period.

Ceiling, disbursement and deadlines — tracked

- ▶ **Against the S\$42m ceiling.** Every submission is measured against the manpower grant cap.
- ▶ **70 / 30 disbursement gate.** Up to 70% disbursed before completion; the final 30% on project completion + T&Cs.
- ▶ **Deadlines computed for you.** 12-month Practitioner's Report, 183-day final claim, progress updates, inspection rights.

EDB Grant Compliance

ST Engineering HR 526-16249-RIS(C)

Claim 01 Jan 2026 – 30 Jun 2026
Support rate confirmed

Simple Advanced New claim

Claimable

13

+1 eligible, \$0 (no hours)

Needs a document

1

Not eligible

5

Total claim

\$492,934.55

Document check

Eligibility

Claim amount

Grant & compliance

Submission pack

How this submission sits against the EDB Support Package's grant ceiling and disbursement gate, plus the audit & reporting obligations.

Maximum grant (manpower only)

\$42m

This claim \$492,934.55 · 1.17% of grant

Disbursable pre-completion (70%)

\$29.4m

This claim is fully disbursable now.

Held back until completion (30%)

\$12.6m

Released on project completion + all T&Cs met.

Audit & reporting obligations

- Each claim covers at least 3 months
This claim window spans 6 month(s) (01 Jan 2026 – 30 Jun 2026).
- External audit by an ACRA-registered Public Accountant (Practitioner)
All claims must be audited under SSRS 4400 and the Practitioner's Report sent directly from the Practitioner to EDB.
- Practitioner's Report at least every 12 months
From the start of the first claim period, with the stamped Claim Forms. due 01 Jan 2027
- Final Audited Claim within 183 days of the Qualifying Period end
Qualifying Period ends 31 Dec 2028; full documentation due by the date shown. due 02 Jul 2029
- Annual Progress Update to EDB
Submitted as and when notified by EDB.
- Final Progress Update within 183 days of project completion
Within 183 calendar days from completion or termination of the Project.
- BA progress report to IHQ — annually, before any claims
All Business Areas submit a progress report to IHQ each year before claim submission.
- Permit EDB inspection on ≥ 2 weeks' written notice
EDB officers (or nominees) may inspect the premises where the Project is carried out.

Support rate 60% (confirmed). Claim window spans 6 month(s); each claim must cover at least 3.

Ask

Upload anything reasonable — the system sorts it out

- ▶ **Guided, in order.** EDB output template and trainee list first, then timesheet / ECMF / payroll — a live tracker ticks each off and shows what's still missing.
- ▶ **Auto-categorised.** Files are recognised by name; a leave report or CPF statement is never mistaken for the core timesheet.
- ▶ **Flexible formats.** Synonym headers ('Emp No', 'Salary'), a Period/Pay-Date column, or one payslip per file with the month in its name — all parse and merge.
- ▶ **Clear reasons when rejected.** If a file can't be read it explains why (missing sheet/column, wrong period) and how to fix it — never a stack trace.

From any figure to its source, in one click

- ▶ **One-click traceability.** Click a number → the exact cell in the original payslip or timesheet, highlighted.
- ▶ **Three outputs, one per reader.** EDB submission template · Statement of Expenditure (auditor) · Issues list (HR).
- ▶ **Built for SSRS 4400.** Designed for the auditor's $\geq 85\%$ value sampling — every claimed dollar is traceable.

basic_salary
/home/stgdx03/Developer/EDB_Grant_Automation/sample_data/payroll.xlsx · sheet Payroll · cell E2

The cited cell is highlighted. Showing a window of the sheet.

	A	B	C	D	E	F	G	H	I
1	Employee ID	Name	Year	Month	Basic Salary	Allowances	Bonus	AWS	CPF (Empl
2	ANS-001	Lim Jia Hao	2026	1	9500	950	0	0	1615
3	ANS-001	Lim Jia Hao	2026	2	9500	950	0	0	1615
4	ANS-001	Lim Jia Hao	2026	3	9500	950	0	0	1615
5	ANS-001	Lim Jia Hao	2026	4	9500	950	0	0	1615
6	ANS-001	Lim Jia Hao	2026	5	9500	950	0	0	1615
7	ANS-001	Lim Jia Hao	2026	6	9500	950	4750	9500	1615
8	ANS-002	Nurul Aisyah Binte Hassan	2026	3	8000	800	0	0	1360
9	ANS-002	Nurul Aisyah Binte Hassan	2026	4	8000	800	0	0	1360
10	ANS-002	Nurul Aisyah Binte Hassan	2026	5	8000	800	0	0	1360
11	ANS-002	Nurul Aisyah Binte Hassan	2026	6	8000	800	4000	8000	1360
12	ANS-003	Tan, Wei Ming	2026	1	7000	700	0	0	1190
13	ANS-003	Tan, Wei Ming	2026	2	7000	700	0	0	1190
14	ANS-003	Tan, Wei Ming	2026	3	7000	700	0	0	1190

The audit explained — SSRS 4400

- ▶ **Agreed-upon procedures — not an audit or a review.** An ACRA-registered Public Accountant (the “Practitioner”) performs a set of procedures agreed with EDB and reports factual findings only — no opinion or assurance is given; EDB draws its own conclusions.
- ▶ **At least 85% of claimed value is sampled.** The Practitioner tests $\geq 85\%$ of the claimed value against the Statement of Expenditure (SOE), vouching each sampled figure to its source document.
- ▶ **Direct to EDB, on a cadence.** The Practitioner’s Report goes straight from the Practitioner to EDB — at least every 12 months, with the final audited claim within 183 days of the qualifying-period end.
- ▶ **What we built for it.** One-click traceability and the SOE output make this engagement fast and clean; outputs stay labelled “UnauditedClaim” until the Practitioner completes their procedures.

Ask the claim anything, in plain English

- ▶ **Natural-language Q&A.** “Why is this person excluded?”, “How is the claim calculated?”, “What's the 9-month rule?”
- ▶ **Recall across claims.** “What documents do we have for E001?” fetches that person's saved document checklist from an earlier claim — with citations — even in a new session.
- ▶ **Local AI.** Powered by a model running on ST Engineering's own DGX — private, no external calls.
- ▶ **Grounded.** It explains and reasons, but never invents a figure — answers cite the source.

The screenshot displays the EDB Grant Compliance interface. At the top, it shows the claim period from 01 Jan 2026 to 30 Jun 2026, with a support rate confirmed. The interface includes tabs for Simple, Advanced, and New claim. A summary section shows: Qualify 14, Needs a document 1, Not eligible 5, and Total claim \$680,614.55. Below this are tabs for Document check, Eligibility, Claim amount, Grant & compliance, and Submission pack. A warning message states: "1 document(s) need attention before those rows can be claimed." The main content area lists two entities: ST Engineering Advanced Networks & Sensors Pte Ltd (with 1 supporting item flagged for audit) and ST Engineering Digital System Pte Ltd. A table for ST Engineering Digital System Pte Ltd shows a missing payslip for Month 4. A chat window titled "Grant & Verification Assistant" is open, showing a question about claim calculation and a detailed response explaining the calculation method.

EDB Grant Compliance
ST Engineering · HR · S26-10249-RIS(C)

Claim 01 Jan 2026 – 30 Jun 2026
Support rate confirmed

Simple Advanced **New claim**

Qualify **14** Needs a document **1** Not eligible **5** Total claim **\$680,614.55**

Document check Eligibility Claim amount Grant & compliance Submission pack

⚠ 1 document(s) need attention before those rows can be claimed.

ST Engineering Advanced Networks & Sensors Pte Ltd
> 1 supporting item(s) flagged for audit (non-blocking)

ST Engineering Digital System Pte Ltd

PERSON	MISSING	DETAIL
DSG-006	Payslip · Month 4	missing April payslip for DSG-006 (Siti Nurhaliza Rahman)

Grant & Verification Assistant
Local model connected

How is each person's claim amount worked out, in simple terms?

For each month, multiply the capped monthly salary by the portion of the month the person was involved and the portion of their full-time effort spent on the project. Sum these monthly amounts across the claim period, then multiply by the support rate. Working days are counted as weekdays at 8.8 hours per day, and only the final total is rounded.

Ask about the claim or a person...

Modern, on-premise, auditable

- ▶ **Web app on a Python engine.** A clean React interface over the deterministic calculation core.
- ▶ **Local AI + local data.** Qwen model on the DGX; a per-employee SQLite store on the machine.
- ▶ **Nothing leaves the building.** All salary and claim data processed on-premise; no internet calls.
- ▶ **Auditable AI.** Every model call is temperature-0, schema-constrained, cached and logged.

POC complete — ready to pilot

- ▶ **Built and verified.** Full pipeline end-to-end on synthetic data; 115 automated tests passing.
- ▶ **Pending confirmation.** Exact support rate & 9-month anchoring (Letter of Award); the auditor's final document list.
- ▶ **Next.** Pilot on a real entity's data; connect to live payroll and timesheet exports; finalise outputs with the appointed auditor.

Thank you

Questions & discussion
